

Phase-Matching in Nonlinear Optical Compounds: A Materials Perspective

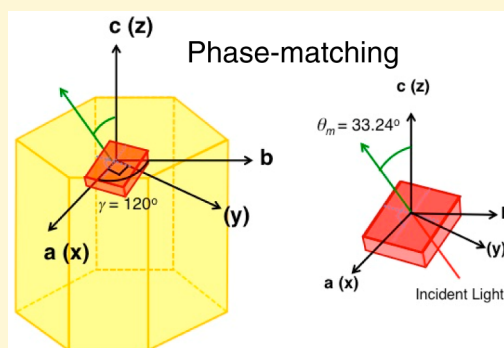
Weiguo Zhang,^{†,¶} Hongwei Yu,^{†,¶} Hongping Wu,^{†,‡} and P. Shiv Halasyamani^{*,†,¶}

[†]Department of Chemistry, University of Houston, 112 Fleming Building, Houston, Texas 77204, United States

[‡]Xinjiang Technical Institute of Physics and Chemistry, 40-1 Beijing Road, Urumqi, Xinjiang 830011, China

S Supporting Information

ABSTRACT: Angle phase-matching in nonlinear optical (NLO) materials is critical for technological applications. The purpose of this manuscript is to describe the concept of phase-matching for the materials synthesis NLO community. Refractive index and birefringence are defined with respect to uniaxial and biaxial crystal systems. The phase-matching angle and wavelength range, Type I and Type II, are explained using real NLO materials, $K_3B_6O_{10}Cl$ (KBOC) and $Ba_3(ZnB_5O_{10})PO_4$ (BZBP). In addition, we describe how refractive index measurements are performed on single crystals and how the resulting birefringence impacts the phase-matching. Our goal is to provide a description of phase-matching that is relevant for the materials synthesis NLO community.



INTRODUCTION

For nonlinear optical (NLO) materials, the ability to phase-match, or be phase-matchable, is of paramount importance. It is through phase-matching that NLO materials are technologically applicable. In other words, it is not enough for a given NLO material to be highly efficient at a particular wavelength; the material must also be phase-matchable. However, what exactly is the role of structure and chemistry in phase-matching? Importantly, how is phase-matching relevant to functional inorganic NLO materials?

The purpose of this manuscript is to

- Describe the concept of phase-matching in powders and crystals, i.e., Type I and Type II. Refractive index, birefringence, uniaxial, and biaxial will also be defined.
- Discuss how the refractive index is measured on crystals.
- Describe how the phase-matching angle and wavelength range for uniaxial and biaxial crystals can be determined.
- Describe how the birefringence of the material impacts the phase-matching angle and wavelength range.

We will be discussing angle phase-matching (sometimes called index matching, angle matching, or birefringence matching), but hereafter simply phase-matching, that is distinct from quasi-phase-matching^{1–3} and noncritical phase-matching methods.^{4,5}

PHASE-MATCHING

The most straightforward definition of phase-matching is when the wave vectors of the fundamental radiation and second-harmonic radiation are equal. This occurs in a second-harmonic generating (SHG) material when the refractive indices of the fundamental wave and second-harmonic wave are equal, i.e., $n(\omega) = n(2\omega)$. In their seminal 1968 paper, Kurtz and Perry,⁶ and

updated 1976 paper, Dougherty and Kurtz,⁷ demonstrated that phase-matching could be determined from sieved powders. (In the 1968 and 1976 papers, the terms Type I and Type II phase-matching are not used. The phase-matching determined from powders is an average value of Type I and Type II.) If the SHG efficiency is measured as a function of particle size, one can determine if the material is phase-matchable or not (see Figure 1). It is important, however, to recognize the limitations of this measurement. A particle size vs SHG intensity measurement:

- tells you if the material is phase-matchable at a specific wavelength, the incident wavelength used.

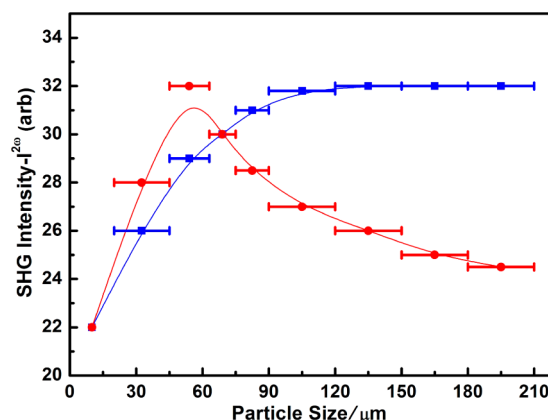


Figure 1. Phase-matching (blue) and nonphase-matching (red) curves for powder SHG measurements.

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- does not provide any information about the phase-matching angle.
- does not provide any information about the phase-matching wavelength range. (In principle, the phase-matching wavelength range can be determined from powders by using a tunable laser source.)

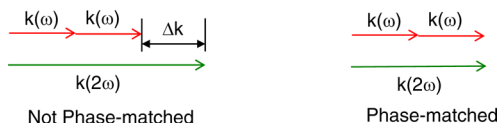
Even with these limitations, the measurement is highly useful and can guide the experimentalist if crystal growth should be pursued. To obtain the phase-matching angle and wavelength range, large single crystals (>5 mm) are needed.

Phase-matching in crystals is more involved owing to anisotropy in the dielectric response and refractive indices. In our discussion, we are going to assume the fundamental wavelength, with angular frequency ω , is 1064 nm; thus, the SHG wavelength, with angular frequency 2ω , is 532 nm. Our discussion, however, is valid for any fundamental and SHG wavelengths.

In the SHG process, there are fundamental and second-harmonic wave vectors, $k(\omega)$ and $k(2\omega)$, respectively, propagating through the crystal. The wave vectors, $k(\omega)$ and $k(2\omega)$, may be defined as⁸

$$k(\omega) = \omega \times n(\omega)/c; \quad k(2\omega) = 2\omega \times n(2\omega)/c$$

with $n(\omega)$ being the refractive index at 1064 nm, $n(2\omega)$ being the refractive index at 532 nm, and c being the speed of light. In general, $n(2\omega) \neq n(\omega)$, and as such, there is a phase-mismatch, Δk , between the two waves, i.e., $\Delta k = k(2\omega) - 2k(\omega) = (4\pi/\lambda)[n(2\omega) - n(\omega)]$.



The most efficient SHG will occur with $\Delta k = 0$. This is known as the phase-matching condition.⁹ In real materials, k is vector quantity, and thus, the direction of propagation in the material where $n(2\omega) = n(\omega)$, i.e., the refractive index of the second-harmonic wave is equal to the refractive index of the fundamental wave, should be considered. The crystalline anisotropy and frequency dispersion of the refractive index, n , requires a discussion of birefringence and refractive index measurements.

■ BIREFRINGENCE

A material's birefringence is the difference in the refractive indices at a specific wavelength and occurs only in anisotropic crystal systems, i.e., noncubic crystal classes. The noncubic crystal classes can be divided into two groups: uniaxial and biaxial.

Uniaxial	Biaxial
Hexagonal	Orthorhombic
Tetragonal	Monoclinic
Trigonal	Triclinic

The unique optic axis in a uniaxial material is along the highest symmetry axis, i.e., C_6 , C_4 , and C_3 for hexagonal, tetragonal, and trigonal crystal classes, respectively. The direction of the optic axis or axes in a biaxial material depends on the refractive indices. For a uniaxial system, a beam of light is split into ordinary (o) and extraordinary (e) beams if its propagation direction is other than along optic axis, i.e., n_o and n_e . The value of the birefringence in a uniaxial optical material is the difference between n_o and n_e , $\Delta n = |n_o - n_e|$ (see Figure 2). For $n_e > n_o$ ($n_e < n_o$) the material is termed a positive (negative) uniaxial crystal.^{10,11}

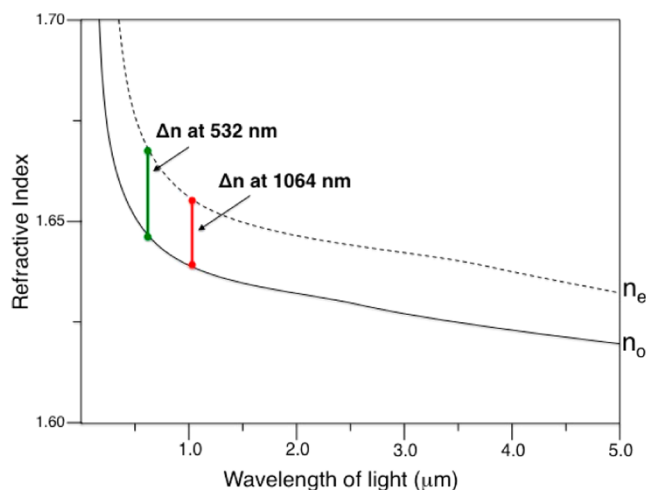


Figure 2. Refractive index as a function of wavelength for a positive uniaxial crystal. The birefringences at 532 and 1064 nm are indicated by the vertical green and red lines, respectively.

Two specific examples, quartz and KH_2PO_4 , are given below. Quartz crystallizes in the trigonal system and exhibits $n_o = 1.54822$ and $n_e = 1.55746$ at 508 nm.¹² It should be noted that the refractive indices change as a function of wavelength. As $n_e > n_o$, quartz is considered a positive uniaxial crystal. Figure 3 shows the

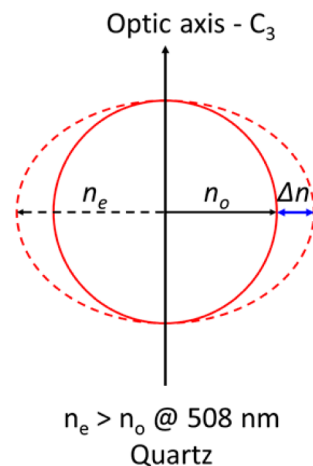


Figure 3. Two-dimensional projection of the refractive index of quartz with $n_e > n_o$ is shown. The birefringence Δn is also indicated.

two-dimensional projection of the refractive index quadric for quartz in the xz -plane. Throughout this paper, n_o and n_e will be shown as solid and dashed lines, respectively. In addition, what is depicted for quartz and for all the examples in this paper is a two-dimensional slice of the three-dimensional optical indicatrix (refractive index tensor). The birefringence, Δn (blue double-headed arrow), is $|n_e - n_o|$.

An example of a negative uniaxial crystal is tetragonal KH_2PO_4 (KDP). At 546 nm, $n_o = 1.51152$ and $n_e = 1.46982$.¹³ The projection of the refractive indices for KDP in the xz -plane is shown in Figure 4. In this situation, since $n_o > n_e$, the solid circle is larger than the dashed ellipse. Again, the birefringence (Δn) is shown as a blue double-headed arrow.

For materials in one of the biaxial crystal systems, the situation is more complex. There are three refractive indices, n_x , n_y , and n_z , along the x -, y -, and z -axis, respectively. By convention, $n_x < n_y < n_z$. If $n_z - n_y > n_y - n_x$ ($n_z - n_y < n_y - n_x$), the material is termed a

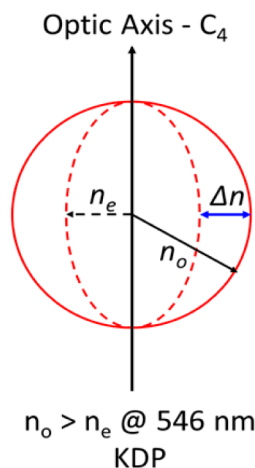


Figure 4. Two-dimensional projection of the refractive index of KDP with $n_o > n_e$ is shown. The birefringence Δn is also indicated.

positive (negative) biaxial optical crystal.^{10,11} The value of the birefringence is taken to be the difference between the maximum values of n_z and n_x (see Figure 5). An example of a positive

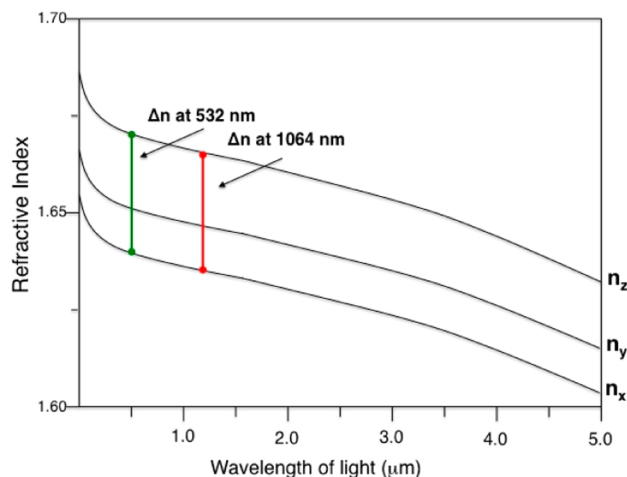


Figure 5. Refractive index as a function of wavelength for a positive biaxial crystal. The birefringences at 532 and 1064 nm are indicated by the green and red vertical lines, respectively.

crystal is KTiOPO_4 (KTP). KTP crystallizes in the orthorhombic system and exhibits $n_x = 1.7381$, $n_y = 1.7458$, and $n_z = 1.8302$ at 1064 nm.¹⁴ The birefringence, Δn , is thus $n_z - n_x$, $1.8302 - 1.7381 = 0.0921$. The projections of KTP's refractive indices in the xz -, xy -, and yz -planes are shown in Figure 6.

Refractive Index Measurements. In order to determine the birefringence, one must measure the refractive index at a variety of wavelengths. One of the most accurate methods for measuring the refractive indices of a solid is through the minimum deviation technique¹⁵ on a large crystal. In this method, a crystal of the material is grown and cut as a wedge or prism, polished, and indexed. Monochromatic sources from the UV to the IR are used to measure the refractive indices at several wavelengths.^{16–35} An example of this was recently reported for $\text{Ba}_3(\text{ZnB}_5\text{O}_{10})\text{PO}_4$ (BZBP).³⁶ The polished and indexed wedge shaped crystal for BZBP is shown in Figure 7. As seen, a relatively large crystal (several mm) is needed for the measurement. In addition, the vertex angle is inversely related to the birefringence; i.e., the smaller the birefringence, the larger the vertex angle

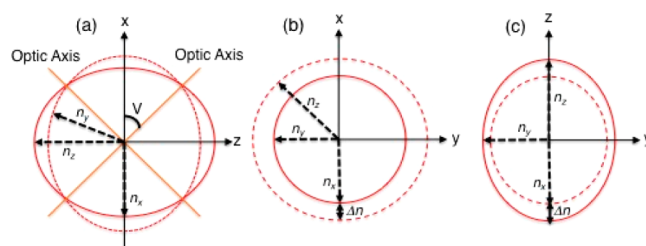


Figure 6. Two-dimensional projection of the refractive indices for KTP in the xz -, xy -, and yz -planes. The “V” in (a) is the acute angle between the z -axis and optic axis.

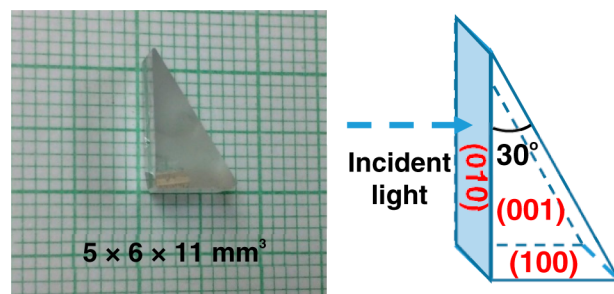


Figure 7. A wedge shaped crystal (left) of BZBP that is used for refractive index measurements by the minimum deviation technique. The indexed faces and vertex angle are shown on the right.

needs to be in order to obtain the refractive index values at a range of wavelengths. Figure 8a shows the refractive index data for BZBP using the minimum deviation technique from the UV to the IR. As BZBP is biaxial, three refractive indices, n_x , n_y , and n_z , are observed. Also since $n_z - n_y < n_y - n_x$, BZBP is a negative biaxial crystal. It is also possible to determine the refractive indices using smaller crystals. A five wavelength, 450.2, 532, 636.5, 829.3, and 1062.6 nm, prism coupler sold by Metricon Co. (<http://www.metricon.com>; no endorsement is implied) enables one to determine the refractive indices at the aforementioned wavelengths using a smaller crystal, 2–4 mm (Figure 9). Again, the crystal needs to be polished and indexed. Using these crystals, the refractive index data for BZBP was also acquired (see Figure 8b). With both measurements, we are able to fit the refractive index data as a function of wavelength to the Sellmeier equation:³⁷

$$n_i^2 = A + \frac{B}{\lambda^2 - C} - D\lambda^2$$

where λ is the wavelength in μm and A , B , C , and D are the Sellmeier parameters. As there are four Sellmeier parameters to be determined, five wavelengths are the minimum required. Thus, the minimum deviation technique is more accurate compared with the prism coupling method. The solid lines in Figures 8a and b are the fits to the refractive index data for BZBP using the Sellmeier equation. Once the Sellmeier equations are known, it is possible to calculate the refractive index at any wavelength, and as we will demonstrate, it allows one to determine the phase-matching wavelength range.

■ PHASE-MATCHING IN UNIAXIAL AND BIAxIAL SYSTEMS

The purpose of the previous sections was to describe birefringence in uniaxial and biaxial systems and refractive index measurements. This information will be used to describe Type I and Type II phase-matching.

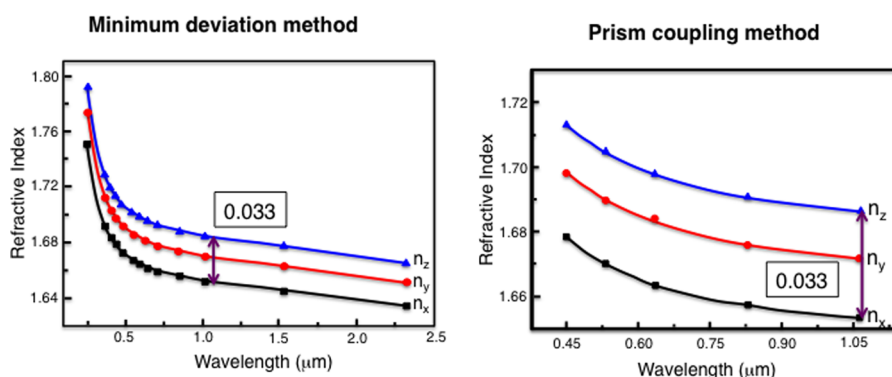


Figure 8. Refractive index data of BZBP by the (a) minimum deviation method and (b) prism coupling method. In both instances, a birefringence of 0.033 at 1064 nm was determined. With both figures, the curves are the fits to the Sellmeier equation.

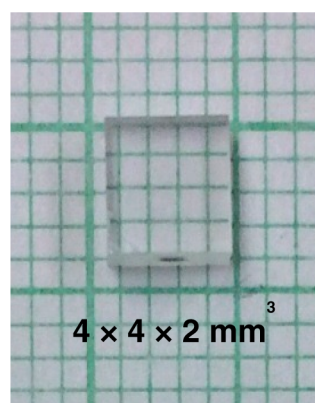


Figure 9. BZBP crystal used for the prism coupling refractive index measurement method.

Uniaxial. Type I Phase-Matching. The projection of the refractive indices in the xz -plane at ω , 1064 nm (red), and 2ω , 532 nm (green), are shown in Figure 10a,b for a positive uniaxial crystal, $n_e > n_o$. A tetragonal crystal system is assumed; thus, the optic axis is parallel to the C_4 axis.

When Figure 10a,b is “combined”, see Figure 10c, the phase-matching angle can be determined. Recall that for phase-matching $n(2\omega) = n(\omega)$. As seen in Figure 10c, this occurs when $n_o(2\omega)$ (solid green circle) intersects with $n_e(\omega)$ (dashed red ellipse), green dot. Another way to describe $n(2\omega) = n(\omega)$ is by $e(\omega) + e(\omega) \rightarrow o(2\omega)$, i.e., two extraordinary waves at the fundamental wavelength combine to produce an ordinary wave at the second-harmonic. The Type I phase-matching angle, $\theta_m(I)$, is taken from the optic axis to the green arrow. The phase-matching “point” in three-dimensions form symmetrical cones (see Figure 11). The phase-matching angle, $\theta_m(I)$, can be calculated from³⁸

$$\sin^2(\theta_m) = \left(\frac{n_e(\omega)}{n_o(2\omega)} \right)^2 \frac{n_o^2(\omega) - n_o^2(2\omega)}{n_o^2(\omega) - n_e^2(\omega)}$$

where refractive indices $n_e(\omega)$, $n_o(\omega)$, and $n_o(2\omega)$ can be determined experimentally as described previously.

For a negative uniaxial crystal (tetragonal), the situation is similar (see Figure 12). Again the solid and dashed red curves represent the ordinary and extraordinary waves, respectively, at 1064 nm, whereas the solid and dashed green curves represent the ordinary and extraordinary waves, respectively, at 532 nm. Recall that for a negative uniaxial crystal $n_o > n_e$. In this situation,

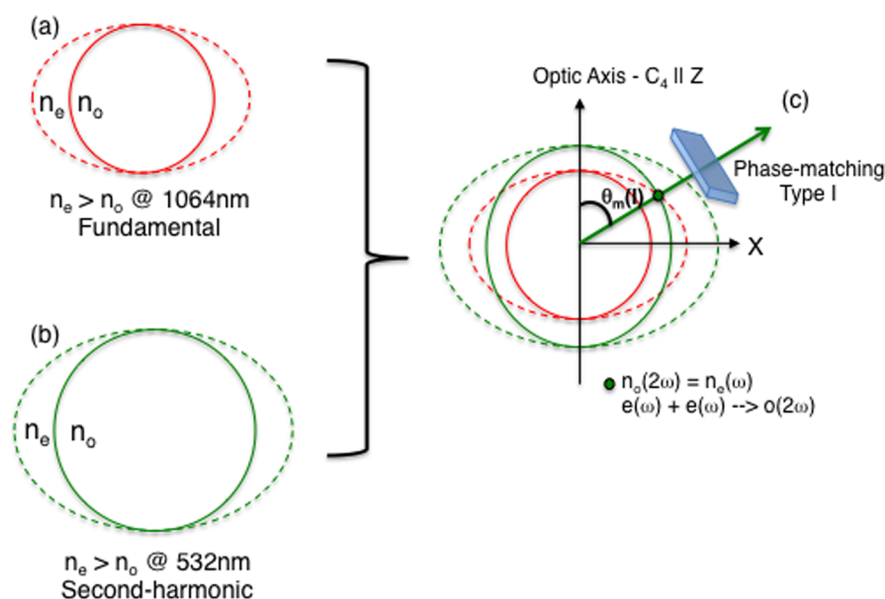


Figure 10. Projections of the refractive indices at (a) ω , 1064 nm, and (b) 2ω , 532 nm, are shown for a *positive uniaxial crystal*. These curves are combined in (c) and a phase-matching angle, $\theta_m(I)$, is determined.

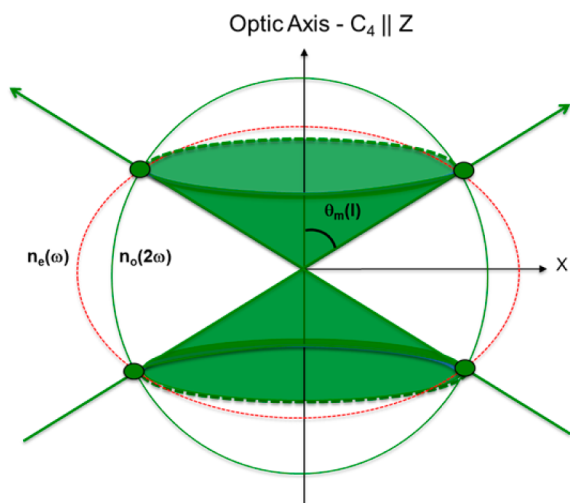


Figure 11. Symmetric cones (green) for Type I phase-matching for positive uniaxial crystals.

$n(2\omega) = n(\omega)$ when $n_e(2\omega) = n_o(\omega)$; i.e., the green dashed ellipse intersects with the solid red circle or $o(\omega) + o(\omega) \rightarrow e(2\omega)$, the green dot. That is two *ordinary* waves at the fundamental wavelength combine to produce an *extraordinary* wave at the second-harmonic.

Similar to the positive uniaxial situation, with negative uniaxial materials symmetrical phase-matching cones are also observed (see Figure 13). In a negative uniaxial crystal, the phase-matching angle, $\theta_m(I)$, can be calculated from³⁸

$$\sin^2(\theta_m) = \left(\frac{n_e(2\omega)}{n_o(\omega)} \right)^2 \frac{n_o^2(2\omega) - n_o^2(\omega)}{n_o^2(2\omega) - n_e^2(2\omega)}$$

where refractive indices $n_e(2\omega)$, $n_o(\omega)$, and $n_o(2\omega)$ can be determined experimentally.

Type II Phase-Matching. In some SHG materials, a second symmetrical phase-matching cone is observed. With Type II

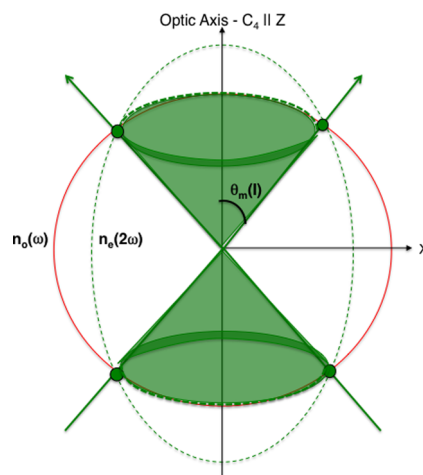


Figure 13. Symmetric cones (green) for Type I phase-matching for negative uniaxial crystals.

phase-matching, phase-matching occurs when $n_e(2\omega) = [n_e(\omega) + n_o(\omega)]/2$ or $n_o(2\omega) = [n_e(\omega) + n_o(\omega)]/2$. The shape of the $[n_e(\omega) + n_o(\omega)]/2$ curve for both the positive and negative uniaxial situations is an ellipse.

Positive Uniaxial: Type II Phase-Matching. For positive uniaxial crystals, Type II phase-matching occurs when $[n_e(\omega) + n_o(\omega)]/2 = n_o(2\omega)$ (see Figure 14). In Figure 14, a tetragonal system is assumed, and as before, the solid and dashed red curves represent the ordinary and extraordinary waves, respectively, at 1064 nm, whereas the solid and dashed green curves represent the ordinary and extraordinary waves, respectively, at 532 nm. With Type II phase-matching, an *extraordinary* and *ordinary* wave from the fundamental wavelength combine to produce an *ordinary* wave at the second-harmonic, $e(\omega) + o(\omega) \rightarrow o(2\omega)$. The calculation of the Type II phase-matching angle is more complicated as $[n_e(\omega) + n_o(\omega)]/2 = n_o(2\omega)$. In this situation,³⁸

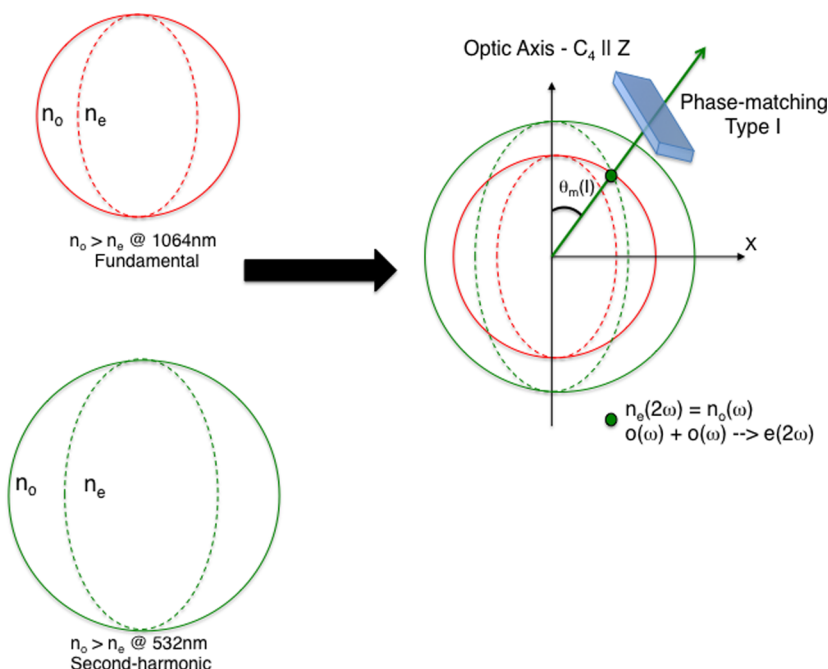


Figure 12. Projections of the refractive indices at (a) 1064 nm and (b) 532 nm are shown for a *negative uniaxial crystal*. These curves are combined in (c) and a phase-matching angle, $\theta_m(I)$, is determined.

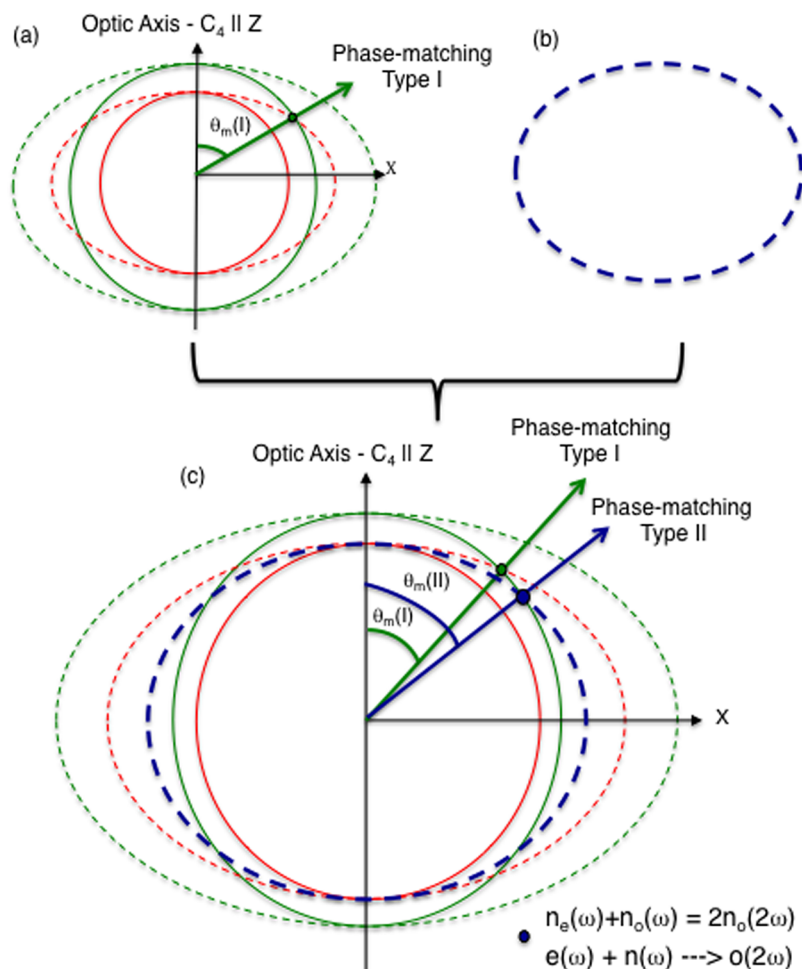


Figure 14. (a) Type I phase-matching for a positive uniaxial crystal. (b) Ellipse representing $[n_e(\omega) + n_o(\omega)]/2$. These curves are combined in (c), and the Type II phase-matching angle, $\theta_m(II)$, is determined. The Type I phase-matching angle, $\theta_m(I)$, is also shown.

$$\frac{1}{2} \left(\frac{n_o(\omega)n_e(\omega)}{\sqrt{n_o^2(\omega)\sin^2\theta_m + n_e^2(\omega)\cos^2\theta_m}} + n_o(\omega) \right) = n_o(2\omega)$$

since $n_o(\omega)$, $n_e(\omega)$, and $n_o(2\omega)$ are known, one can numerically solve this nonlinear equation for θ_m .

Negative Uniaxial: Type II Phase-Matching. For negative uniaxial crystals, Type II phase-matching occurs when $[n_e(\omega) + n_o(\omega)]/2 = n_e(2\omega)$ (see Figure 15). That is, an extraordinary and ordinary wave from the fundamental wavelength combine to produce an extraordinary wave at the second-harmonic, $e(\omega) + o(\omega) \rightarrow e(2\omega)$.

Similar to the positive uniaxial situation, calculating theta for Type II phase-matching in negative uniaxial crystal is possible. One can determine $\theta_m(II)$ by solving the following equation given that $n_o(\omega)$, $n_e(\omega)$, $n_o(2\omega)$, and $n_e(2\omega)$ are known.³⁸

$$\frac{1}{2} \left(\frac{n_o(\omega)n_e(\omega)}{\sqrt{n_o^2(\omega)\sin^2\theta_m + n_e^2(\omega)\cos^2\theta_m}} + n_o(\omega) \right) = \frac{n_o(2\omega)n_e(2\omega)}{\sqrt{n_o^2(2\omega)\sin^2\theta_m + n_e^2(2\omega)\cos^2\theta_m}}$$

As with Type I phase-matching, for Type II phase-matching, symmetrical cones are observed. These cones (blue) along with the Type I phase-matching cones (green) are shown in Figure 16.

A summary of the phase-matching conditions and angles for uniaxial systems is given in Table 1.

Biaxial. With biaxial systems, the situation is more cumbersome compared with uniaxial crystal classes owing to three independent refractive indices from which phase matching can be achieved. Technically, the phase-matching condition for biaxial crystals is similar to uniaxial crystals, i.e., the intersection of the refractive index surfaces at the fundamental and second-harmonic frequencies, $n(\omega) = n(2\omega)$. When light travels in a biaxial crystal along any direction other than the optic axis, the beam is split in two, each of which has different refractive indices. The beam with the smaller refractive index is termed fast (f), whereas the beam with the larger refractive index is termed slow (s). These slow and fast beams from the fundamental light combine to generate the second-harmonic radiation. This process is shown schematically in Figure 17.

Figure 17a,b shows the refractive index curves at 1064 and 532 nm, respectively. Recall that, for biaxial crystal systems, there are three refractive indices, n_x , n_y , and n_z . Figure 17a,b is combined in Figure 17c, and as in the uniaxial situation, phase-matching occurs when $n(\omega) = n(2\omega)$. Type I phase-matching occurs when two fundamental slow beams, $n_\omega(s)$, combine to generate a second-harmonic beam, $n(2\omega)(f)$, i.e., where the solid red, $n(\omega)$, and solid green, $n(2\omega)$ curves cross. Type II phase-matching occurs when a fundamental slow, $n_\omega(s)$, and fundamental fast, $n_\omega(f)$, combine to generate a second-harmonic

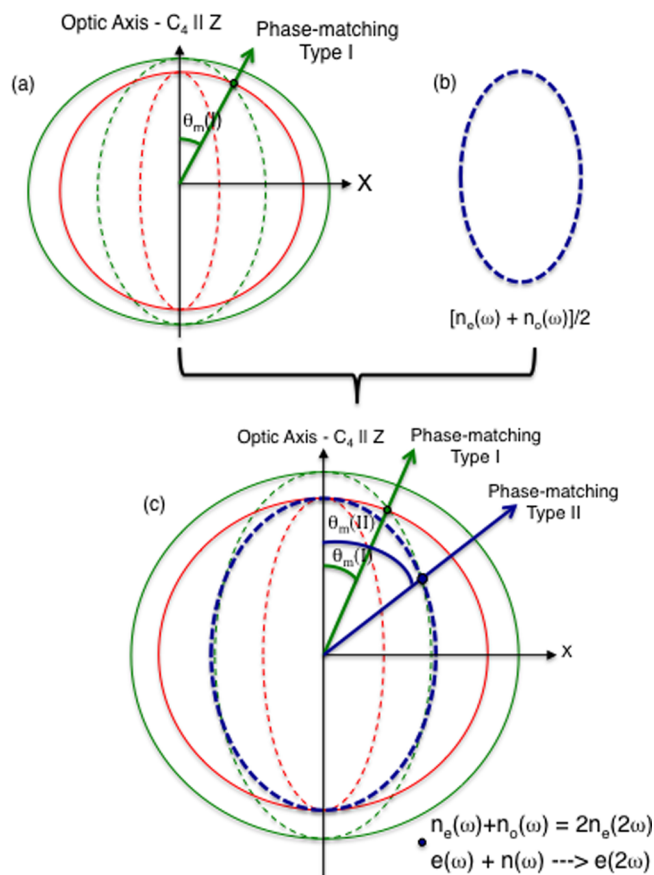


Figure 15. (a) Type I phase-matching for a negative uniaxial crystal. (b) Ellipse representing $[n_e(\omega) + n_o(\omega)]/2$. These curves are combined in (c), and the Type II phase-matching angle, $\theta_m(II)$, is determined.

beam, $n(2\omega)(f)$, i.e., where the blue dashed curve ($n_\omega(s) + n_\omega(f))/2$ crosses the solid green curve, $n(2\omega)$.

The phase-matching conditions for biaxial systems has been detailed extensively by Hobden.³⁸ He determined 13 phase-matching conditions for biaxial crystals. The list of these conditions and phase-matching loci are beyond the scope of this paper.

Calculating the phase-matching angles, θ_m and ϕ_m , in biaxial systems is more complicated compared with uniaxial systems. One must solve the following equation for θ and ϕ .

$$\frac{k_x^2}{n^{-2} - n_x^{-2}} + \frac{k_y^2}{n^{-2} - n_y^{-2}} + \frac{k_z^2}{n^{-2} - n_z^{-2}} = 0$$

where $k_x = \sin \theta \cos \phi$, $k_y = \sin \theta \sin \phi$, and $k_z = \cos \theta$. θ is the angle from the z-axis, and ϕ is the angle from the x-axis in the xy-plane. The three principle refractive indices are n_x , n_y , and n_z . Explicit solutions for the above equation for Type I and Type II phase-matching in biaxial systems are given by Yao and Fahlen³⁹ and are deposited in the Supporting Information (S1).

Additional Comments About Type I and Type II Phase-Matching. Is there a preference between Type I and Type II phase-matching? No, either may be used for NLO applications. As the NLO efficiency is wavelength dependent, there may be situations where Type I is preferred over Type II or vice versa.

Is the Type II phase-matching wavelength range always smaller than the Type I? Yes, for Type I phase-matching, the wavelength is in the range of $n_{\max}(\omega) > n_{\min}(2\omega)$, whereas for Type II the range of $[n(\omega) + n(\omega)]/2 > n_{\min}(2\omega)$ (Please see Figures 25 and 26). As $(n_o(\omega) + n_e(\omega))/2$ is never larger than $n_o(\omega)$

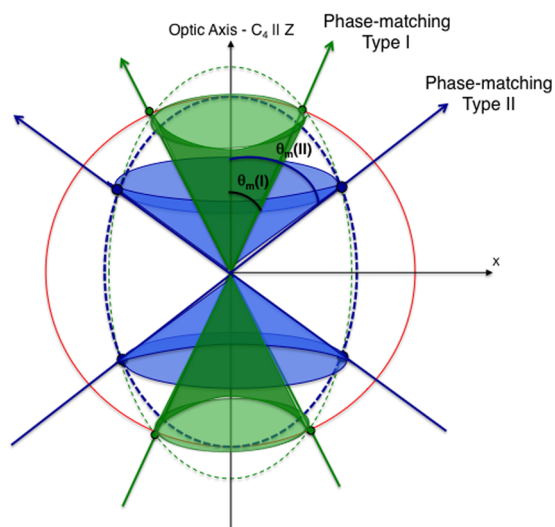


Figure 16. Negative uniaxial phase-matching cones for Type I, green, and Type II, blue, are shown. The respective phase-matching angles, $\theta_m(I)$ and $\theta_m(II)$, are also given.

(negative situation) or $n_e(\omega)$ (positive situation), the phase-matching wavelength range of Type II will be always smaller than Type I. This is true for both negative and positive uniaxial and biaxial crystals.

Does Type II phase-matching always occur? No, if a material is Type I phase-matchable, it may be Type II phase-matchable, or it may not. However, if a material is Type II phase-matchable, it is always Type I phase-matchable.

■ PHASE-MATCHING ANGLE AND WAVELENGTH RANGE

In addition to the phase-matching angles, θ_m (uniaxial) and θ_m and ϕ_m (biaxial), there are also phase-matching wavelength ranges, which arise from the dispersion of the refractive indices with wavelength. These are the wavelength ranges where phase-matching is possible, and there are distinct wavelength ranges for Type I and Type II phase-matching.

We will describe specific uniaxial and biaxial examples to describe the phase-matching wavelength ranges. As a uniaxial example, we will use $K_3B_6O_{10}Cl$ (KBOC), a positive uniaxial NLO material.^{34,40} KBOC was reported in 2010 as a SHG material with a deep UV absorption edge, 180 nm.⁴⁰ The material exhibits a powder SHG efficiency of $4 \times$ KDP at 1064 nm and was shown to be phase-matchable. A large crystal of KBOC was grown by top-seeded solution growth methods using a KF/KCl/ PbO/H_3BO_3 flux.³⁴ A large crystal of KBOC was cut into a wedge, polished, and indexed (Figure 18), and the refractive indices were measured by the minimum deviation method at several wavelengths between 200 and 2000 nm. The birefringence is 0.0464 at 1064 nm (Figure 19). The fits to the Sellmeier equations are also shown in Figure 19. Since the Sellmeier equations are known, we can calculate n_o and n_e at any wavelength of our choosing. By using the Type I and Type II positive uniaxial equations given previously, we can calculate θ_m as a function of wavelength, λ . This data is shown in Figure 20. Figure 20 provides us with a great deal of information. First, we can determine the fundamental wavelength ranges for Type I and Type II phase-matching by extrapolating the black and red curves to the y-axis. KBOC is Type I phase-matchable from 544 to 3772 nm and Type II phase-matchable from 748 to 2848 nm.

Table 1. Phase-Matching Conditions and Phase-Matching Angle Calculations for Uniaxial Systems

	Uniaxial	
	Phase-Matching Condition	Phase-Matching Angle (θ_m)
Type I, positive	$n_o(2\omega) = n_e(\omega)$ $e(\omega) + e(\omega) \rightarrow o(2\omega)$	$\sin^2(\theta_m) = \left(\frac{n_e(\omega)}{n_o(2\omega)} \right)^2 \frac{n_o^2(\omega) - n_o^2(2\omega)}{n_o^2(\omega) - n_e^2(\omega)}$
Type I, negative	$n_e(2\omega) = n_o(\omega)$ $o(\omega) + o(\omega) \rightarrow e(2\omega)$	$\sin^2(\theta_m) = \left(\frac{n_e(2\omega)}{n_o(\omega)} \right)^2 \frac{n_o^2(2\omega) - n_o^2(\omega)}{n_o^2(2\omega) - n_e^2(2\omega)}$
Type II, positive	$[n_e(\omega) + n_o(\omega)]/2 = n_o(2\omega)$ $e(\omega) + o(\omega) \rightarrow o(2\omega)$	$\frac{1}{2} \left(\frac{n_o(\omega)n_e(\omega)}{\sqrt{n_o^2(\omega)\sin^2\theta_m + n_e^2(\omega)\cos^2\theta_m}} + n_o(\omega) \right) = n_o(2\omega)$
Type II, negative	$[n_e(\omega) + n_o(\omega)]/2 = n_e(2\omega)$ $e(\omega) + o(\omega) \rightarrow e(2\omega)$	$\frac{1}{2} \left(\frac{n_o(\omega)n_e(\omega)}{\sqrt{n_o^2(\omega)\sin^2\theta_m + n_e^2(\omega)\cos^2\theta_m}} + n_o(\omega) \right) = \frac{n_o(2\omega)n_e(2\omega)}{\sqrt{n_o^2(2\omega)\sin^2\theta_m + n_e^2(2\omega)\cos^2\theta_m}}$

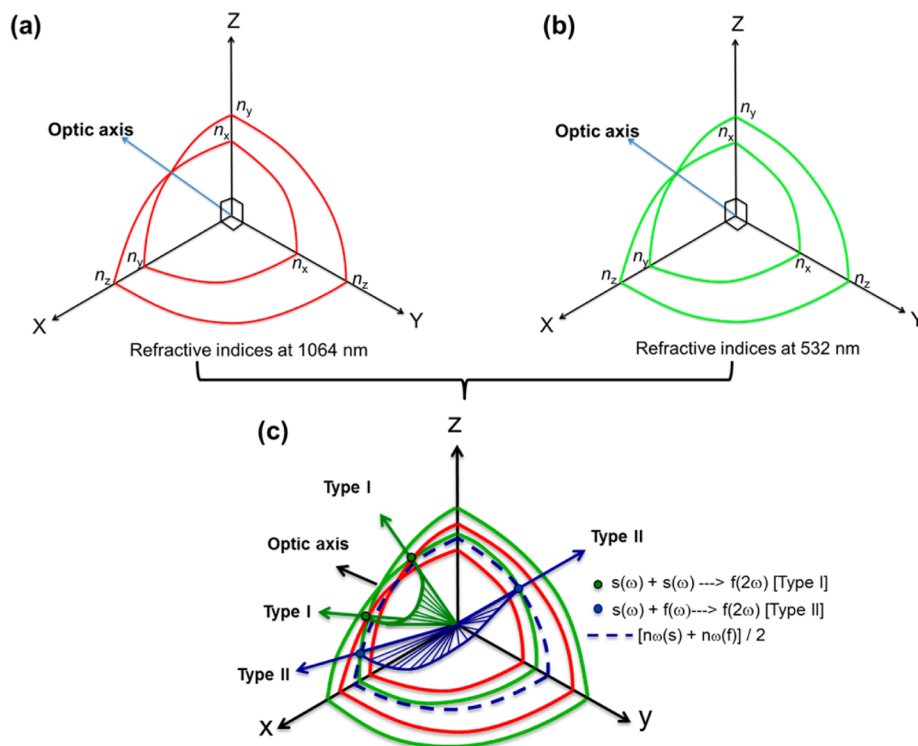


Figure 17. Refractive indices for a biaxial crystal at (a) 532 nm and (b) 1064 nm. These curves are combined in (c), and the Type I (green dots) and Type II (blue dots) phase-matching conditions are indicated.

Figure 20 also informs us of the exact angle to cut the crystal for phase-matching at a specific wavelength. The black squares in Figure 20 represent 1064 nm. For Type I phase-matching, the propagation direction is 33.24° with respect to the optic axis (z-axis) and is perpendicular to the crystal wafer, whereas for Type II phase-matching the propagation direction is 51° with respect to the optic axis. Both the values are read off the x-axis in Figure 20.

How the Type I phase-matching angle relates to an ideal crystal is shown in Figure 21 (yellow crystal). In KBOC, the crystallographic axes, a , b , and c , are not fully coincident with the

principle optical axes x , y , and z . As stated earlier, a Type I phase-matching angle of 33.24° is determined (Figure 20) and is the angle from the optic axis (z-axis). A crystal wafer is cut at this phase-matching angle, and phase-matching achieved when the crystal is placed perpendicular to the propagation direction of the incident beam. For Type II phase-matching, the crystal would be cut at 51° with respect to the optic axis.

With a biaxial NLO material, we will use $\text{Ba}_3(\text{ZnB}_5\text{O}_{10})\text{PO}_4$ (BZBP) as an example.^{36,41} This material was reported in 2015 as both a polycrystalline powder and large centimeter size crystals.

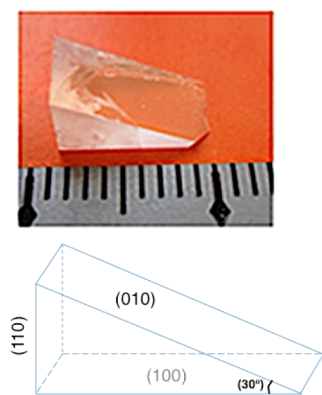


Figure 18. Wedge shaped crystal (top) of KBOC. The indexed faces and vertex angle are given in the bottom figure.

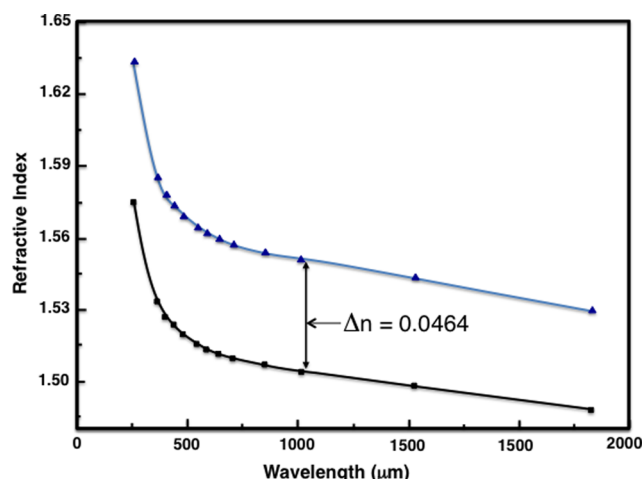


Figure 19. Refractive index data for KBOC using the minimum deviation method is shown. A birefringence of 0.0464 at 1064 nm is determined. The curves are fits from the Sellmeier equation.

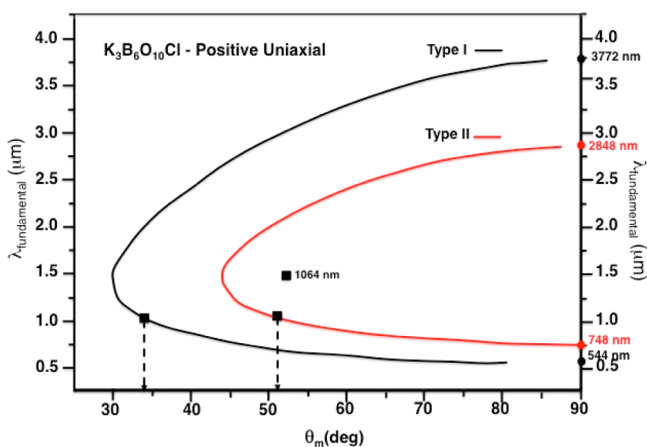


Figure 20. Phase-matching angle, θ_m , with respect to the fundamental wavelength, λ . The black and red curves represent the Type I and Type I phase-matching angle and wavelength ranges. The Type I (Type II) phase-matching wavelength ranges are given by the black (red) circles on the right y-axis.

The powder SHG efficiency was shown to be $4 \times$ KDP at 1064 nm, and an absorption edge of 180 nm was measured. We have already shown, in Figure 7, the polished and indexed wedge shaped crystals, the refractive index data, and fits to the Sellmeier

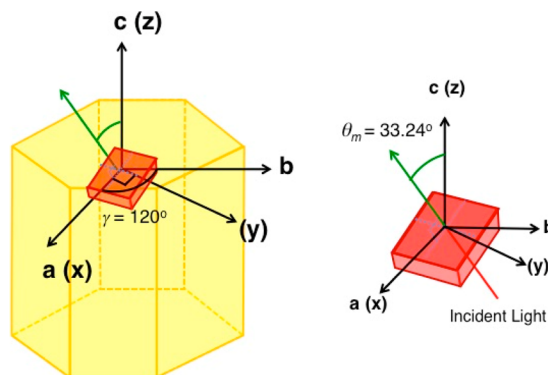


Figure 21. An ideal KBOC crystal is shown on the left in yellow. A Type I phase-matching angle of 33.24° has been determined. A crystal wafer is cut, shown in red (right), at the phase-matching angle.

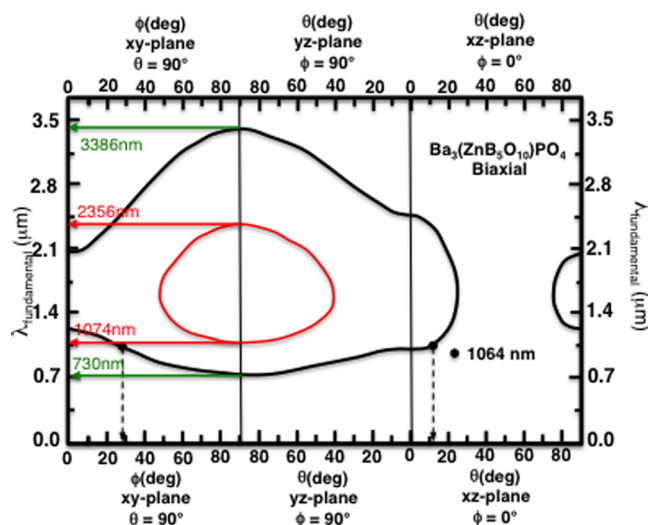


Figure 22. Phase-matching angles in principle planes, θ and ϕ , with respect to the fundamental wavelength for BZBP. The Type I (Type II) phase-matching curves are shown in black (red). The Type I (Type II) phase-matching wavelength range is 730–3386 nm (1074–2356 nm).

equations, respectively, for BZBP. With a biaxial crystal, there are two phase-matching angles, θ_m and ϕ_m . θ_m represents the phase-matching angle with respect to the z -axis, whereas ϕ_m represents the phase-matching angle in the xy -plane with respect to the x -axis. The equations necessary to solve for θ_m and ϕ_m with respect to the fundamental wavelength are given in the Supporting Information (S2). For BZBP, the phase-matching angles (θ_m , ϕ_m) with respect to the fundamental wavelength are shown in Figure 22. As with the uniaxial situation, it is possible to determine the Type I and Type II phase-matching wavelength ranges by extrapolating the maxima of the black and red curves, respectively, to the y -axis. Thus, for BZBP, the Type I (Type II) phase-matching angle range is 730–3386 nm (1074–2356 nm). A second figure, θ_m vs ϕ_m , is needed to determine the angles necessary for phase-matching. This is shown in Figure 23 for 1064 nm. The dashed line in Figure 23 informs us of the θ_m and ϕ_m angles needed for Type I phase-matching at 1064 nm. The crystal may be cut at any point along this curve with the corresponding θ_m and ϕ_m angles to achieve phase-matching. An ideal crystal example should be illustrative.

An ideal crystal of BZBP is shown in Figure 24 (yellow crystal). From Figure 23, we observe Type I phase-matching is achieved

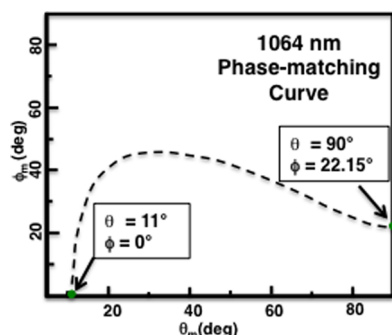


Figure 23. Phase-matching angles, θ_m and ϕ_m , for BZBP at 1064 nm.

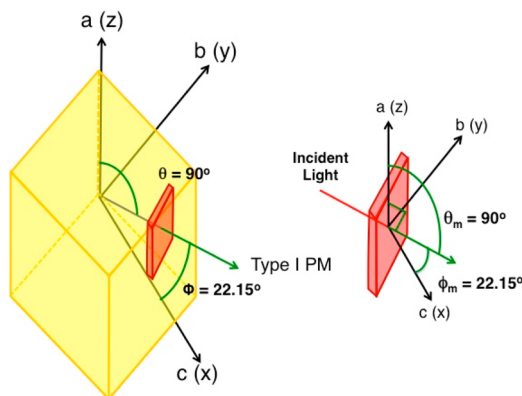


Figure 24. An ideal BZBP crystal is shown in yellow (left). A crystal wafer, shown in red, is cut such that $\theta_m = 90^\circ$ and $\phi_m = 22.15^\circ$. When the crystal is cut at these phase-matching angles and placed perpendicular to the propagation direction of the incident beam, phase-matching is achieved.

when $\theta = 90^\circ$ and $\phi = 22.15^\circ$. Recall that θ is the angle from the optic axis (z -axis), whereas ϕ is the angle in the xy -plane. A crystal wafer, shown in red, is cut at these phase-matching angles, and phase-matching is achieved when the crystal is placed perpendicular to the propagation direction of the incident beam.

It is also possible to determine the Type I and Type II phase-matching wavelength ranges, but not the phase-matching angles, through the Sellmeier equations. We will again use KBOC and BZBP as examples.

The refractive indices as a function of wavelength, along with the Sellmeier equation fits and birefringence at 1064 nm for KBOC, are given in Figure 19. As we observed in Figure 20, the Type I and Type II phase-matching wavelength ranges for fundamental light are 544–3722 and 748–2848 nm, respectively. With the Sellmeier equations that were determined from the refractive index measurements, we are able to calculate the refractive index at any wavelength. Figure 25 shows the calculated refractive indices as a function of fundamental and second-harmonic wavelengths for KBOC. Recall that KBOC is uniaxial, so the n_o and n_e at ω and 2ω may be calculated. These are shown as red solid and dashed green lines, respectively, in Figure 25. The dashed blue line represents $(n_o(\omega) + n_e(\omega))/2$. Recall that phase-matching occurs when $n(\omega) = n(2\omega)$. As seen in Figure 25, this occurs when the dashed red line, $n(\omega)$, intersects with the solid green line, $n(2\omega)$. These intersections are shown by blue circles in Figure 25. Drawing a vertical line from the blue circles to the bottom x -axis indicates the Type I phase-matching range. As seen in Figure 25, and consistent with Figure 20, the Type I phase-matching range for the fundamental wavelength is

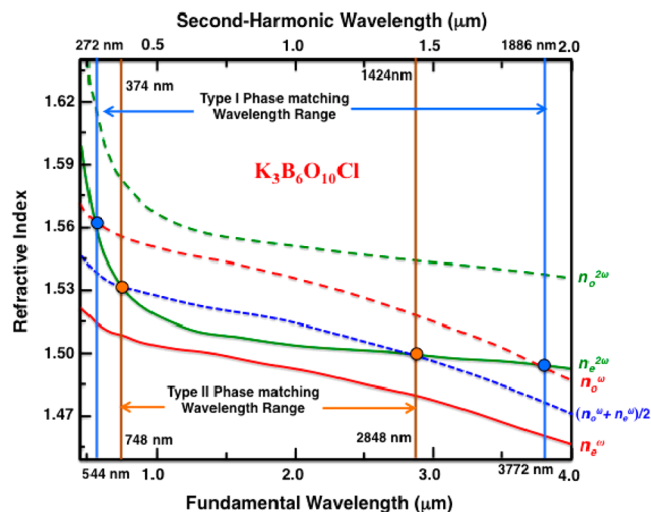


Figure 25. Calculated refractive indices of the ordinary and extraordinary wavelengths for the fundamental (red) and second-harmonic (green) wavelengths using the Sellmeier equations. The blue dashed line represents $(n_o(\omega) + n_e(\omega))/2$. The blue and orange circles represent the Type I and Type II phase-matching wavelength ranges, respectively.

from 544 to 3722 nm. Type II phase-matching occurs when $2n_e(2\omega) = n_o(\omega) + n_e(\omega)$ or when $n_e(2\omega) = (n_o(\omega) + n_e(\omega))/2$. In Figure 25, this is when the dashed blue line, $n_e(2\omega)$, intersects the solid green line, $(n_o(\omega) + n_e(\omega))/2$. These intersections are shown by orange circles in Figure 25. The Type II phase-matching wavelength range can be determined by drawing vertical lines from the orange circles to the bottom x -axis. For KBOC, the Type II phase-matching wavelength range is from 748 to 2848 nm.

Analogous data can be used for biaxial systems. For BZBP, the calculated refractive indices as a function of wavelength are shown in Figure 26. Again, the refractive indices at the funda-

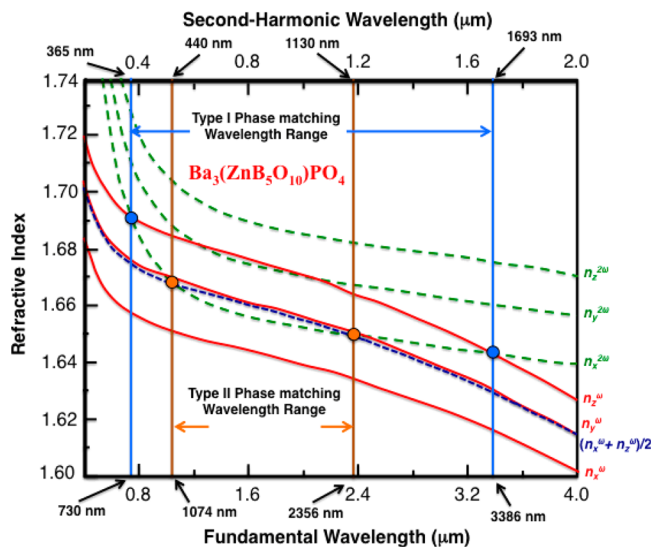


Figure 26. Calculated refractive indices of the ordinary and extraordinary wavelengths for the fundamental (red) and second-harmonic (green) wavelengths for BZBP using the Sellmeier equations. The blue dashed line represents $(n_x(\omega) + n_z(\omega))/2$. The blue and orange circles represent the Type I and Type II phase-matching wavelength ranges, respectively.

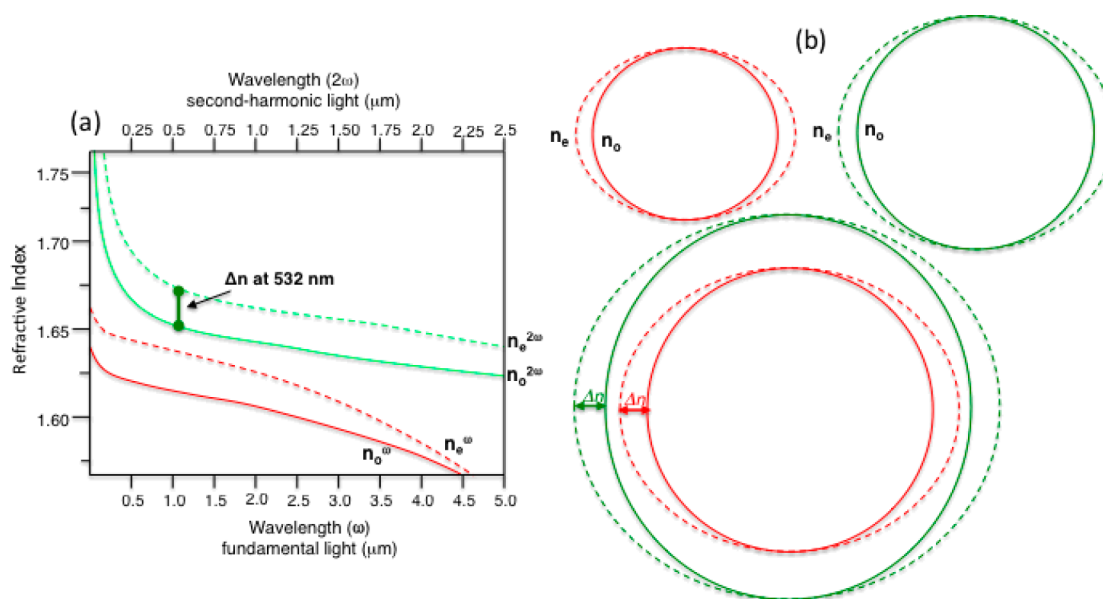


Figure 27. (a) Refractive indices as a function of wavelength with the shorter and longer wavelengths shown in green and red, respectively. Note that the green and red curves do not cross at any point. (b) Projection of the refractive indices at 532 nm (green) and 1064 nm (red). The bottom figure shows that the green and red circles and ellipses do not intersect.

menatal (second-harmonic) wavelengths are shown as red and dashed green lines, respectively. In addition, $(n_x(\omega) + n_z(\omega))/2$ is shown as a blue dashed line. As with the uniaxial situation, Type I phase-matching occurs when $n(\omega) = n(2\omega)$. In Figure 26, this occurs when the solid red line, $n(\omega)$, intersects with the dashed green line, $n(2\omega)$. These intersections are shown by blue circles and represent the extent of the Type I phase-matching wavelength region, i.e., from 730 to 3386 nm. Type II phase-matching occurs when $2n_x(2\omega) = n_x(\omega) + n_z(\omega)$ or when $n_x(2\omega) = [n_x(\omega) + n_z(\omega)]/2$, i.e., when the dashed green line intersects with the dashed blue line. These are shown by the orange circles and represent the Type II phase-matching wavelength range, i.e., from 1074 to 2356 nm. Thus, for both uniaxial and biaxial crystals, once the refractive indices are measured and the Sellmeier equations determined, one can ascertain the Type I and Type II phase-matching wavelength ranges.

■ RELATIONSHIPS BETWEEN PHASE-MATCHING AND BIREFRINGENCE

This section will describe how the phase-matching wavelength range is critically dependent on the birefringence.

Birefringence Too Small: Phase-Matching Does Not Occur. In materials that are nonphase-matchable, $n(\omega) = n(2\omega)$ is never achieved. This can be understood through Figure 27a,b. In both panels, a positive uniaxial crystal is assumed. Recall that phase-matching occurs when $\Delta k = 0$, which is when $n(\omega) = n(2\omega)$. As seen in Figure 27a, the birefringence is too small and the $n(2\omega)$ curve (solid green line) never crosses $n(\omega)$ (dashed red line), and as such, phase-matching never occurs. This can also be understood through a two-dimensional projection of the refractive indices (see Figure 27b). In this figure, it is also observed that the $n(\omega)$ and $n(2\omega)$ curves do not intersect since the birefringence is too small.

Increase in Birefringence Increases Phase-Matching Wavelength Range. The phase-matching wavelength will increase as the birefringence increases. With the following discussion, it is assumed that phase-matching (Type I) occurs, i.e., $n(\omega) = n(2\omega)$. In Figure 28a, a uniaxial system is assumed

and the birefringence at 532 nm is indicated by a green vertical line. Phase-matching is observed when $n(2\omega)$ crosses $n(\omega)$, i.e., when the solid green curve crosses the dashed red curve, indicated by the two blue circles. The Type I phase-matching wavelength range (second harmonic) is between these blue circles, vertical blue lines, and is from approximately 400 nm to 1.5 μm (top x -axis). As the birefringence increases (from Figure 28a,b), the Type I phase-matching wavelength range is much larger, between approximately 200 nm and 2.15 μm . A similar situation is observed with a biaxial system (see Figure 28c,d). Here, the Type I phase-matching wavelength range (second harmonic) increases from approximately 400–2000 to 200–2400 nm as the birefringence increases.

Birefringence Too Large: Walk-Off Effects. It should not be assumed from the previous section that the larger the birefringence the better the NLO properties. If the birefringence becomes too large, serious walk-off effects can occur.^{42,43} This is schematically shown in Figure 29. In Figure 29a, the Type I phase-matching conditions for a negative uniaxial crystal is shown. For these conditions, only the second-harmonic light has a walk-off effect. Figure 29b shows the second-harmonic, 532 nm, ordinary (solid-circle) and extraordinary (dashed ellipse) curves. The walk-off angle, ρ , for Δn is shown in the right of the figure where the k -vector represents the wave-vector, and the S -vector represents the intensity of the second-harmonic beam. The angle between the two is defined as the walk-off angle. As the birefringence increases, $\Delta n'$ (blue) (Figure 29c), this angle, ρ (blue), now between k and S' , increases and the intensity of second-harmonic beam decreases. As such, too large birefringence results in a reduction of the second-harmonic intensity. This is the situation between $\text{CsLiB}_6\text{O}_{10}$ (CLBO) and $\beta\text{-BaB}_2\text{O}_4$ ($\beta\text{-BBO}$). Both materials are excellent SHG materials and are used to generate 266 nm. Even though $\beta\text{-BBO}$ has larger SHG coefficients (see Table 2), CLBO generates greater SHG intensity.⁴⁴ This is attributable to the smaller birefringence of CLBO and thus smaller walk-off angle compared with $\beta\text{-BBO}$.

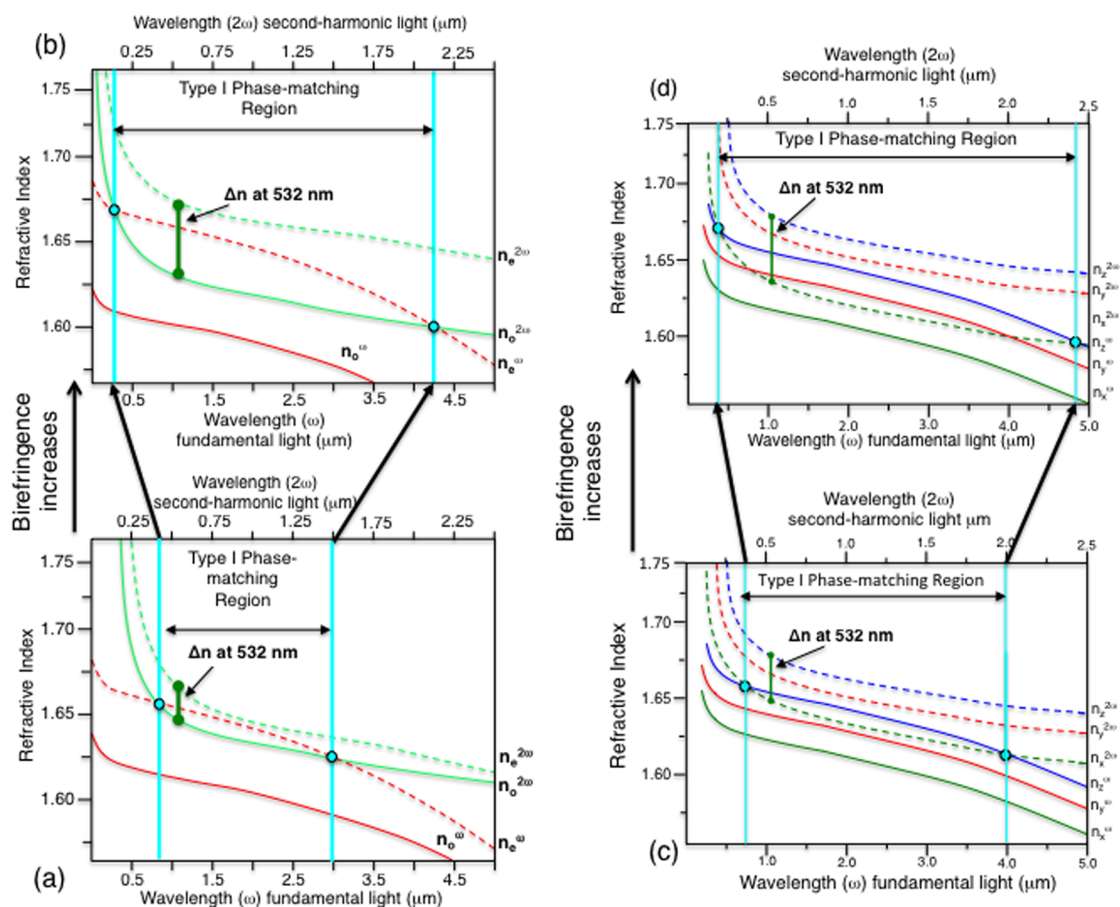


Figure 28. (a) Refractive indices as a function of wavelength at 532 nm (green) and 1064 nm (red) in a positive uniaxial crystal. The Type I phase-matching region is between the intersection of $n_o(2\omega)$ and $n_o(\omega)$. As the birefringence increases in (b), the Type I phase-matching

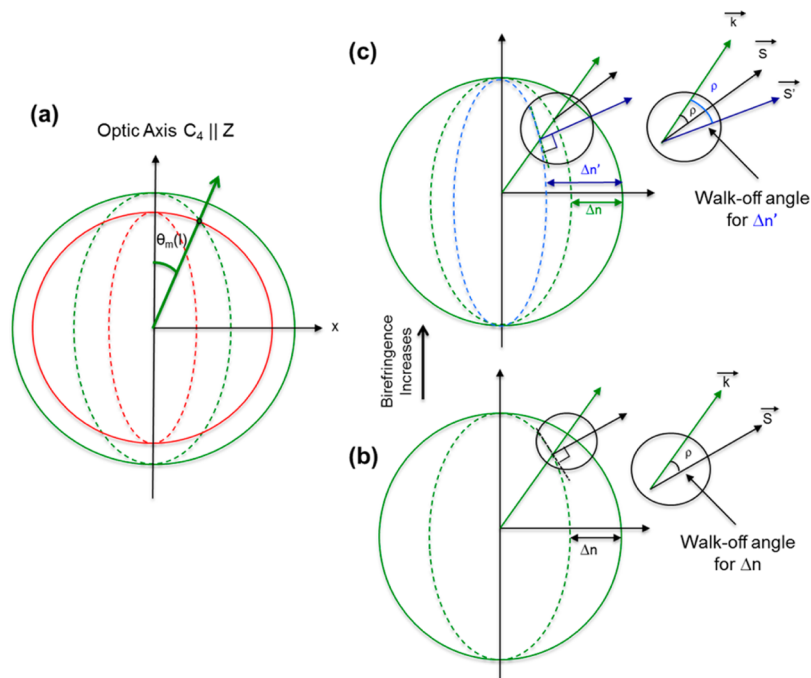


Figure 29. In (a), the Type I phase-matching conditions for a negative uniaxial crystal is shown. The walk-off angle, ρ , is shown in (b). As the birefringence increases, the walk-off angle increases (c) and the SHG intensity decreases.

Table 2. Birefringence (Δn), Walk-Off Angle, and NLO Coefficients (d_{ij}) of CsLiB₆O₁₀ (CLBO) and β -BaB₂O₄ (β -BBO) Crystals

material	Δn (@1064 nm)	walk-off angle (deg)	d_{ij} (pm/V)
CsLiB ₆ O ₁₀ (CLBO)	0.049 ⁴⁵	2.0 ⁴⁵	$d_{36} = 0.92$ ⁴⁶
β -BaB ₂ O ₄ (β -BBO)	0.113 ⁴⁷	3.2 ⁴⁵	$d_{22} = 2.20$ ⁴⁵

CONCLUSION

Phase-matching, or to be phase-matchable at a specific wavelength, is required if NLO material is to have any technological applications. Although phase-matching capabilities can be determined through a powder SHG measurement, i.e., particle size vs SHG efficiency, it is only through the growth of large, high quality single crystals (multimillimeter) that the full phase-matching capabilities, angle and wavelength range, may be determined. Our aim with this manuscript is not only to describe phase-matching, Type I and Type II, in uniaxial and biaxial systems, but also to explain how the refractive indices and birefringence are measured and how these impact the phase-matching wavelengths and angles. To summarize the main points of this paper:

- Powder SHG measurements (particle size vs SHG intensity) are a necessary first step. These measurements inform the experimentalist if the material is phase-matchable at the incident wavelength used.
- Large single crystals, several mm in each dimension, that have been indexed, cut, and polished are needed to determine the refractive indices at several wavelengths and, subsequently, the birefringence.
- The refractive index data as a function of wavelength may be fit to the Sellmeier equation: $n_i^2 = A + \frac{B}{\lambda^2 - C} - D\lambda^2$. Once the Sellmeier equations are known, it is possible to calculate the refractive index at any wavelength.
- With uniaxial crystal classes (hexagonal, tetragonal, and trigonal), the Type I and Type II phase-matching conditions are relatively straightforward, whereas in biaxial systems (orthorhombic, monoclinic, and triclinic), the situation is more complex as phase-matching may be achieved through three independent refractive indices.
- With the Sellmeier equations, one can calculate the phase-matching angle, θ_m , as a function of wavelength for both Type I and Type II phase-matching.
- Also, with the Sellmeier equations, one can plot the wavelength as a function of refractive index and determine graphically the Type I and Type II phase-matching wavelength ranges.

We hope that this paper is useful for the materials synthesis NLO community and provides a better understanding of the phase-matching concept for the experimentalist.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.chemmater.7b00243.

Refractive index expressions for biaxial crystals with respect to θ and ϕ ; the Sellmeier equations for BZBP (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: psh@uh.edu.

ORCID

P. Shiv Halasyamani: 0000-0003-1787-1040

Author Contributions

[¶]W.Z. and H.Y. contributed equally.

Notes

The authors declare no competing financial interest.

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